

Scalability & Future Plan

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| Date          | 30 September 2026 |
| Team ID       | [Your Team ID]    |
| Project Name  | AI BlogNest       |
| Maximum Marks | 1 Mark            |

Scalability & Future Plan:

This document outlines how the AI BlogNest solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

| S.No | Limitation                                                                | Impact                                                  | Priority to Address |
|------|---------------------------------------------------------------------------|---------------------------------------------------------|---------------------|
| 1    | Single Express server instance, no clustering/load balancer               | Limits concurrent request throughput under heavy load   | High                |
| 2    | No caching layer for frequently accessed blogs                            | Repeated GET /api/blogs queries hit PostgreSQL directly | Medium              |
| 3    | AI generation is synchronous and blocks the request until Gemini responds | 3-5 second response times for AI endpoints under load   | Medium              |

Scalability Plan:

| S.No | Scalability Aspect | Current State                                | Proposed Upgrade / Solution                                                                           |
|------|--------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Single Node.js process on localhost:5000     | Deploy behind a load balancer with multiple Node.js instances (PM2 cluster mode / horizontal scaling) |
| 2    | Data Storage       | Single PostgreSQL instance                   | Add read replicas and connection pooling (PgBouncer) as blog/user volume grows                        |
| 3    | Performance        | No caching; direct DB reads on every request | Introduce Redis caching for popular blog reads and rate-limit AI endpoints                            |
| 4    | Security           | JWT + bcrypt, single "registered user" role  | Add refresh tokens, role-based access control, and per-user rate limiting                             |

Future Roadmap:

| Phase   | Planned Feature / Enhancement                                        | Target Timeline | Expected Impact                                                |
|---------|----------------------------------------------------------------------|-----------------|----------------------------------------------------------------|
| Phase 2 | Role-based access control (Admin, Editor, Reader) and comment system | Q1 2027         | Enables collaborative content moderation and richer engagement |
| Phase 3 | Redis caching layer and pagination for blog listing                  | Q2 2027         | Improves response times and supports larger datasets           |
| Phase 4 | Asynchronous AI job queue (BullMQ) for generation/summarization      | Q3 2027         | Removes blocking AI calls and improves throughput under load   |